MULTIPLICITY ONE ON SUBSCHEME CYCLES: COEFFICIENT ONE, SMOOTHNESS AND NON-CONTAINMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original section 0AZA: sources/stacks-chow/tags.txt, source lines 13818–13818. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1I: sources/stacks-chow/tags.txt, source lines 13896–13896. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NQ: sources/stacks-chow/tags.txt, source lines 868–868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NP: sources/stacks-chow/tags.txt, source lines 867–867. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00KW: sources/stacks-chow/tags.txt, source lines 766–766. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety. Let $r, s \geq 0$ and let $Y, Z \subset X$ be closed subschemes with $\dim(Y) \leq r$ and $\dim(Z) \leq s$. Assume $[Y]_r = \sum n_i [Y_i]$ and $[Z]_s = \sum m_j [Z_j]$ intersect properly. Let T be an irreducible component of $Y_{i_0} \cap Z_{j_0}$ for some i_0 and j_0 and assume that the multiplicity (in the sense of Section 0AZA) of T in the closed subscheme $Y \cap Z$ is 1. Then*

1. *the coefficient of T in $[Y]_r \cdot [Z]_s$ is 1,*
2. *Y and Z are nonsingular at the generic point of T,*
3. *$n_{i_0} = 1$, $m_{j_0} = 1$, and*
4. *T is not contained in Y_i or Z_j for $i \neq i_0$ and $j \neq j_0$.*

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Proof. Set $n = \dim(X)$, $a = n - r$, $b = n - s$. Observe that $\dim(T) = r + s - n = n - a - b$ by the assumption that the intersections are proper. Let $(A, \mathfrak{m}, \kappa) = (\mathcal{O}_{X,\xi}, \mathfrak{m}_\xi, \kappa(\xi))$ where $\xi \in T$ is the generic point. Then $\dim(A) = a + b$, see Varieties, Lemma 0A21. Let $I_0, I, J_0, J \subset A$ cut out the trace of Y_{i_0}, Y, Z_{j_0}, Z in $\mathrm{Spec}(A)$. Then $\dim(A/I) = \dim(A/I_0) = b$ and $\dim(A/J) = \dim(A/J_0) = a$ by the same reference. Set $\overline{I} = I + \mathfrak{m}^2/\mathfrak{m}^2$. Then $I \subset I_0 \subset \mathfrak{m}$ and $J \subset J_0 \subset \mathfrak{m}$ and $I + J = \mathfrak{m}$. By Lemma 0B1I and its proof we see that $I_0 = (f_1, \ldots, f_a)$ and $J_0 = (g_1, \ldots, g_b)$ where $f_1, \ldots, g_b$ is a regular system of parameters for the regular local ring A. Since $I + J = \mathfrak{m}$, the map

$$I \oplus J \to \mathfrak{m}/\mathfrak{m}^2 = \kappa f_1 \oplus \ldots \oplus \kappa f_a \oplus \kappa g_1 \oplus \ldots \oplus \kappa g_b$$

is surjective. We conclude that we can find $f_1', \ldots, f_a' \in I$ and $g_1', \ldots, g_b' \in J$ whose residue classes in $\mathfrak{m}/\mathfrak{m}^2$ are equal to the residue classes of $f_1, \ldots, f_a$ and $g_1, \ldots, g_b$. Then $f_1', \ldots, g_b'$ is a regular system of parameters of A. By Algebra, Lemma 00NQ we find that $A/(f_1', \ldots, f_a')$ is a regular local ring of dimension b. Thus any nontrivial quotient of $A/(f_1', \ldots, f_a')$ has strictly smaller dimension (Algebra, Lemmas 00NP and 00KW). Hence $I = (f_1', \ldots, f_a') = I_0$. By symmetry $J = J_0$. This proves (2), (3), and (4). Finally, the coefficient of T in $[Y]_r \cdot [Z]_s$ is the coefficient of T in $Y_{i_0} \cdot Z_{j_0}$ which is 1 by Lemma 0B1I. $\square$